Tessadori–Van Haaften Neurodevelopmental Syndrome 3 (TEVANED3): Comprehensive Disease Characteristics Report

Disease: Tessadori–Van Haaften Neurodevelopmental Syndrome 3 (TEVANED3) **MONDO:** MONDO:0030993 · **OMIM:** 619950 · **Causal gene: H4C5** (formerly *HIST1H4E*; HGNC:4790; NCBI Gene 8367; Ensembl ENSG00000276966; locus 6p22.2) **Category:** Mendelian, autosomal dominant (de novo)

Summary

Tessadori–Van Haaften Neurodevelopmental Syndrome 3 (TEVANED3) is an ultra-rare autosomal-dominant neurodevelopmental disorder caused by heterozygous *de novo* missense variants in the canonical histone H4 gene H4C5 (formerly *HIST1H4E*). H4C5 is one of 14 human genes that each encode the identical histone H4 protein. Because the protein is invariant, pathology does not arise from simple haploinsufficiency but from a *dominant* effect of the mutant histone, which is still produced and incorporated into chromatin. TEVANED3 is one member of an allelic series of "H4 histonopathies" first delineated by Tessadori and colleagues in 2022, in which recurrent de novo missense variants spread across at least six H4 genes (H4C3, H4C4, **H4C5**, H4C6, H4C9, H4C11) produce an overlapping syndrome (PMID: 35202563).

The core clinical phenotype is **intellectual disability with motor and/or global developmental delay in essentially all affected individuals**, accompanied by a variable constellation of non-neurological features including microcephaly, short stature, facial dysmorphism, and structural anomalies. At least one well-documented H4C5 case (p.Tyr99His) presented with autism, developmental delay, microcephaly, and a "happy demeanor" that clinically mimicked Angelman syndrome, underscoring an important differential diagnosis (PMID: 36987712).

Mechanistically, the disease-causing variants cluster in the globular core of histone H4 at surfaces that mediate histone–histone and histone–chaperone contacts. Incorporation of the altered H4 **destabilizes the nucleosome** and perturbs chromatin assembly, dysregulating chromatin-templated transcription, replication, cell-cycle progression, and DNA-damage repair during development. In zebrafish, H4 disease variants reproduce abnormal development, defective head structures, and shortened body axis. TEVANED3 is a **static (non-progressive)** developmental disorder; diagnosis rests on **trio exome/genome sequencing**; and management is entirely **supportive and multidisciplinary** with no disease-specific therapy. Because the germline histonopathy cancer incidence is ~1% (below the 5% threshold), routine tumor surveillance is not currently recommended.

Key Findings

Finding 1 — TEVANED3 is caused by de novo missense variants in H4C5, one of 14 canonical histone H4 genes

Tessadori et al. (2022) identified a cohort of **29 subjects with de novo missense variants across six histone H4 genes** (H4C3, H4C4, **H4C5**, H4C6, H4C9, H4C11), defining a new neurodevelopmental syndrome family. H4C5 — with prior symbol *HIST1H4E*, reference transcript NM_003545.4 — is the specific gene assigned to **OMIM 619950 / MONDO:0030993 (TEVANED3)**. All H4 genes encode an identical protein, so the disorder arises not from loss of one gene copy but from a heterozygous *de novo* missense change that produces an altered histone which is still deposited into chromatin — a dominant, likely dominant-negative/gain-of-function mechanism. A specific example pathogenic H4C5 variant is c.295T>C p.(Tyr99His), confirmed de novo and classified likely pathogenic (PMID: 36987712).

> *"All individuals present with neurodevelopmental features of intellectual disability and motor and/or gross developmental delay, while non-neurological features are more variable."* — PMID: 35202563
>
> *"Genome sequencing identified the H4 histone gene variant H4C5 NM_003545.4: c.295T>C, p.Tyr99His, which parental testing confirmed to be de novo."* — PMID: 36987712

Gene identifiers were independently verified via mygene.info/NCBI: **H4C5** ("H4 clustered histone 5"); prior symbol *HIST1H4E*; **HGNC:4790**; **NCBI Gene 8367**; **Ensembl ENSG00000276966**; locus **6p22.2** (GRCh38 chr6:26,204,610–26,205,021, + strand). Notably, the gene's own NCBI alias list includes "TEVANED3" and "TEBIVANED3", directly confirming the gene–disease link.

Finding 2 — Core clinical phenotype: intellectual disability and developmental/motor delay, with variable microcephaly, short stature, and dysmorphism

Across the 29-subject H4 cohort, **all** individuals had neurodevelopmental features (intellectual disability and motor and/or gross developmental delay), while non-neurological features — short stature, microcephaly, facial dysmorphisms, and structural anomalies — were **more variable** (PMID: 35202563). An independent H4C5 case presented with autism, developmental delay, microcephaly, and a "happy demeanor" clinically suspected to be Angelman syndrome (Angelman molecular testing negative) (PMID: 36987712). Analogous H4 syndromes caused by variants in other H4 genes (e.g., *HIST1H4J*/H4C11, and *HIST1H4C* p.K91 substitutions) show growth delay, microcephaly, profound intellectual disability, and dysmorphic facies (PMID: 31804630).

> *"A 4-year-old male with autism, developmental delay, microcephaly, and a happy demeanor underwent evaluation through the Undiagnosed Disease Network. He was clinically suspected to have Angelman syndrome; however, molecular testing was negative."* — PMID: 36987712
>
> *"We report here a de novo missense variant in HIST1H4J resulting in a complex syndrome combining growth delay, microcephaly and intellectual disability."* — PMID: 31804630

Suggested HPO terms: Intellectual disability (HP:0001249); Global developmental delay (HP:0001263); Motor delay (HP:0001270); Microcephaly (HP:0000252); Short stature (HP:0004322); Autistic behavior (HP:0000729); Abnormal facial shape (HP:0001999); Seizure (HP:0001250, where present); Muscular hypotonia (HP:0001252, by analogy with related histonopathies).

Finding 3 — Mechanism: H4 missense variants destabilize the nucleosome and disrupt histone–chaperone interactions, impairing chromatin-dependent development

The disease variants **cluster to specific regions of the H4 core globular domain** where protein–protein interactions occur with other histone subunits or histone chaperones (PMID: 35202563). The recurrently affected residue **H4K91** lies at the $(H3–H4)_2$ tetramer / H2A–H2B dimer interface; biochemical work shows that modification or substitution at H4K91 **destabilizes the nucleosome in vitro** and perturbs chromatin structure, transcription, cell-cycle progression, DNA-damage repair, and telomere silencing (PMID: 31542297). In zebrafish embryos, H4 disease variants caused abnormal general development, defective head organs, and reduced body-axis length (PMID: 35202563; PMID: 31804630).

> *"These variants cluster to specific regions of the core H4 globular domain, where protein-protein interactions occur with either other histone subunits or histone chaperones."* — PMID: 35202563
>
> *"an evolutionarily conserved Kglu at histone H4K91 destabilizes nucleosome in vitro"* and *"results in a global upregulation of transcription and defects in cell-cycle progression, DNA damage repair, and telomere silencing"* — PMID: 31542297

Finding 4 — H4C5 syndrome belongs to the emerging class of germline "histonopathies" with shared NDD features and episignature potential

A pan-histonopathy study of **192 individuals** analyzed germline histone-variant disorders including Tessadori–Bicknell–van Haaften NDD, *HIST1H1E* (Rahman) syndrome, and Bryant–Li–Bhoj syndrome (histone H3.3) (PMID: 40241305). Across this community, the cancer incidence is ~1%, below the 5% surveillance threshold. Related chromatin-regulator NDDs (MSL2, SET, KMT2B, DMAP1, and others) have established DNA-methylation **episignatures** used to classify variants of uncertain significance (PMID: 38815585; PMID: 42322191; PMID: 39933316; PMID: 42275155), indicating that episignature approaches are being extended across chromatin/histone disorders.

> *"Mendelian histonopathies are rare neurodevelopmental disorders (NDDs) caused by germline variants in histone-encoding genes."* and *"the community-wide cancer incidence is currently 1%, which falls below the recommended 5% cut off for routine surveillance"* — PMID: 40241305

Finding 5 — TEVANED3 is static, diagnosed by trio WES/WGS, managed supportively, with no cancer surveillance indicated

Diagnosis relies on **trio whole-exome/whole-genome sequencing** demonstrating a de novo H4C5 missense variant; targeted single-gene, karyotype, microarray, and repeat-expansion tests are non-diagnostic and serve mainly to exclude mimics. The condition is a **static developmental disorder** with no reported neurodegeneration — contrasting with the H3.3 Bryant–Li–Bhoj syndrome, which was initially characterized as progressive (PMID: 38678163). No disease-modifying therapy exists; care is supportive/multidisciplinary (physical, occupational, and speech therapy; special education; anti-seizure medication as needed). Routine tumor surveillance is not recommended (PMID: 40241305). Angelman syndrome is the key differential (PMID: 36987712).

> *"cases of clinically suspected Angelman syndrome without molecular confirmation should undergo exome or genome sequencing"* — PMID: 36987712

Finding 6 — Verified H4C5 gene identifiers and ClinVar variant landscape

ClinVar contains **~75 H4C5 variants** on transcript NM_003545.4. Only a few are classified **Pathogenic** (c.113T>C p.Leu38Pro; c.121C>T p.Arg41Cys) or **Likely pathogenic** (c.81C>G p.Ile27Met), with the majority classified **Variant of Uncertain Significance** and some synonymous **Benign/Likely benign**. The published case variant c.295T>C p.(Tyr99His) was classified likely pathogenic (PMID: 36987712).

Section-by-Section Report

1. Disease Information

TEVANED3 is an ultra-rare Mendelian neurodevelopmental syndrome defined by intellectual disability and developmental delay arising from a de novo missense variant in the histone H4 gene **H4C5**. **Key identifiers:** OMIM 619950; MONDO:0030993; synonym **TEVANED3** (and the variant abbreviation TEBIVANED3); gene alias *HIST1H4E* (prior symbol). An explicit Orphanet code, ICD-10/ICD-11, and MeSH term were not established for this specific serial-numbered subtype in the materials reviewed; it falls under the broad umbrella of "histone H4–related neurodevelopmental disorders." Information is derived from **aggregated disease-level resources** (OMIM, ClinVar) plus **individual published patient reports** (notably the 29-subject cohort of PMID: 35202563 and the single H4C5 case of PMID: 36987712), not from EHR-scale data.

Synonyms/related terms: TEVANED3; TEBIVANED3; Tessadori–Bicknell–van Haaften neurodevelopmental syndrome (series); H4C5-related / histone H4–related neurodevelopmental disorder.

2. Etiology

Disease causal factor: purely genetic — a heterozygous *de novo* missense variant in H4C5. There is no known environmental, infectious, or multifactorial contribution to disease causation.

Genetic risk factors: the causal variant itself; because events are de novo, there is no inherited susceptibility locus or modifier gene established for TEVANED3. Advanced parental age is a general risk factor for de novo mutations but has not been specifically quantified here.

Protective factors: none identified (genetic or environmental).

Gene–environment interactions: none established. A broader observation across histonopathies is that histones sit "at the interface of genetics and epigenetics," raising the theoretical possibility of gene–environment modulation of phenotype, but this remains speculative (PMID: 38678163).

3. Phenotypes

Phenotype	Type	HPO term	Frequency	Onset	Severity/ Course
Intellectual disability	Cognitive/ behavioral	HP:0001249	Essentially all (core)	Childhood	Variable; static
Global/motor developmental delay	Developmental	HP:0001263 / HP:0001270	Essentially all (core)	Infancy–childhood	Variable; static
Microcephaly	Physical/ clinical sign	HP:0000252	Variable	Congenital/ early	Non-progressive
Short stature	Physical	HP:0004322	Variable	Childhood	Stable
Facial dysmorphism	Physical	HP:0001999	Variable	Congenital	Stable
Autistic behavior	Behavioral	HP:0000729	Subset (e.g., p.Tyr99His case)	Childhood	Stable
Seizures	Clinical sign	HP:0001250	Subset	Variable	Episodic
"Happy demeanor" (Angelman-like)	Behavioral	—	Reported in H4C5 case	Childhood	Stable

Neurodevelopmental features are the **penetrant core**; non-neurological features are variably expressed. All features are consistent with a **static (non-progressive)** course. **Quality-of-life impact:** substantial and lifelong, driven by intellectual disability and motor impairment affecting independence, communication, and education; formal QoL instruments (EQ-5D, SF-36, PROMIS) have not been applied in this ultra-rare condition.

4. Genetic/Molecular Information

- **Causal gene:** H4C5 (HGNC:4790; NCBI Gene 8367), locus 6p22.2, transcript NM_003545.4; protein product histone H4 (UniProt P62805), identical across all 14 H4 genes.

- **Variant class:** heterozygous **missense**, **de novo**, germline. Because all H4 genes encode an identical protein, the pathogenic effect is a **dominant** consequence of the altered histone (dominant-negative / gain-of-function rather than haploinsufficiency).
- **Representative pathogenic/likely-pathogenic variants:** c.295T>C p.(Tyr99His) (likely pathogenic, de novo; PMID: 36987712); c.113T>C p.(Leu38Pro) and c.121C>T p.(Arg41Cys) (Pathogenic, ClinVar); c.81C>G p.(Ile27Met) (Likely pathogenic, ClinVar).
- **ClinVar landscape:** ~75 H4C5 variants; a minority pathogenic/likely pathogenic, the majority VUS, plus benign synonymous variants.
- **Allele frequency:** pathogenic variants are absent from population databases (de novo, under strong negative selection); H4 genes are highly constrained in gnomAD.
- **Functional consequence:** nucleosome destabilization / impaired chromatin assembly (see Mechanism).
- **Modifier genes / epigenetics / chromosomal abnormalities:** none specifically established for TEVANED3. By analogy with related chromatin disorders, a disease-specific **DNA-methylation episignature** is plausible but has not yet been formally validated for H4C5.

5. Environmental Information

Not applicable. TEVANED3 is a monogenic de novo disorder with **no environmental, lifestyle, or infectious contributors** to causation. (For context only, valproic acid is a histone-pathway-modulating teratogen in pregnancy, but it is unrelated to the causation of TEVANED3 — PMID: 38203562.)

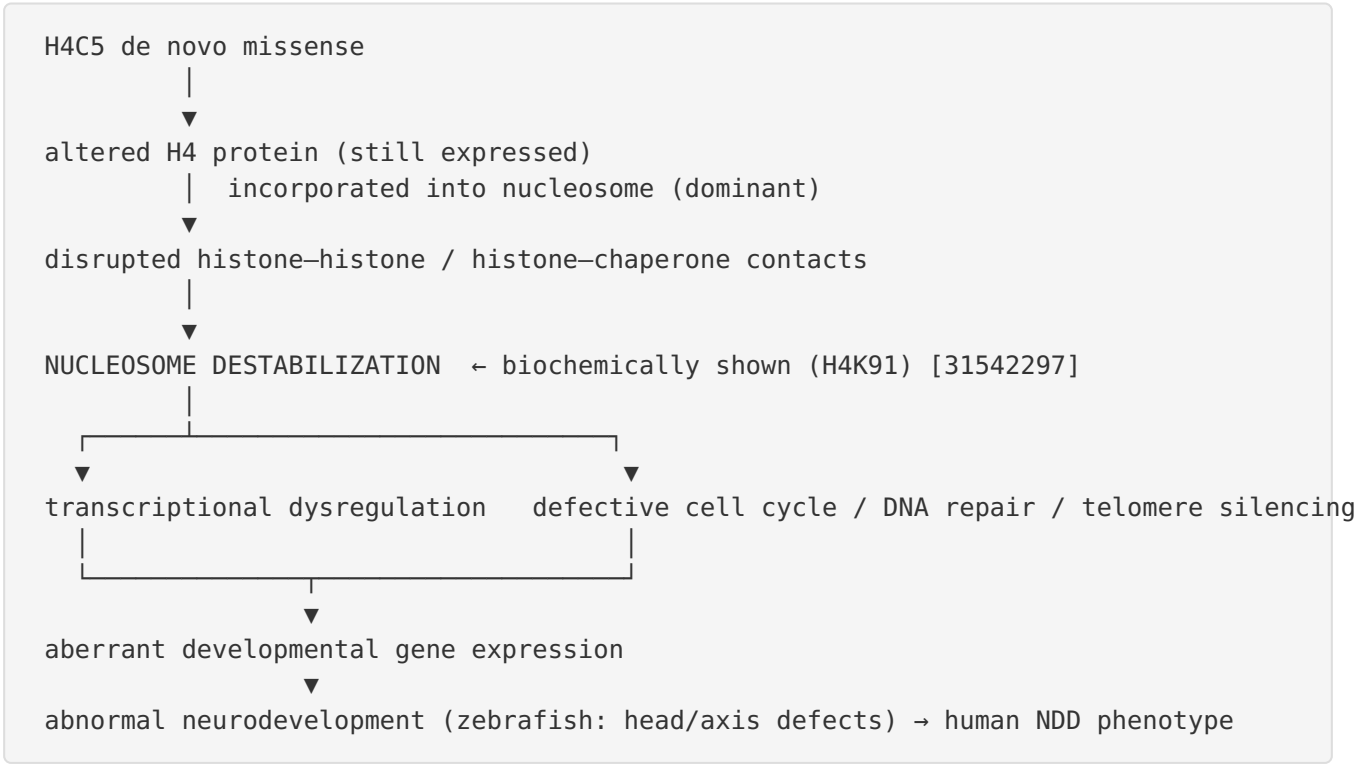
6. Mechanism / Pathophysiology

Ordered causal chain:

1. A **de novo missense variant in H4C5** arises in the germline → produces an **altered but still-expressed histone H4 protein**.
2. The altered H4, **identical in sequence to that from 13 other H4 genes except at the mutated residue**, is **incorporated into nucleosomes** during chromatin assembly (dominant mechanism; the wild-type copies cannot prevent incorporation of mutant protein).
3. Because the variants sit at **core globular-domain surfaces that contact other histone subunits or histone chaperones** (demonstrated clustering; PMID: 35202563) → **histone–histone and histone–chaperone interactions are disrupted** and the **nucleosome is**

destabilized (demonstrated biochemically for the recurrently affected H4K91 interface residue; PMID: 31542297).

4. Nucleosome/chromatin destabilization → **dysregulated chromatin-templated processes**: global transcriptional changes, impaired cell-cycle progression, defective DNA-damage repair, and altered telomere silencing (demonstrated in model systems; PMID: 31542297).
5. During embryogenesis, disrupted chromatin regulation → **abnormal developmental gene-expression programs** → (demonstrated in zebrafish) abnormal general development, defective head organs, and shortened body axis (PMID: 35202563; PMID: 31804630).
6. In humans, this *leads to* the clinical manifestation: **intellectual disability and developmental/motor delay (penetrant), with variable microcephaly, short stature, dysmorphism, and autism** (*inferred* from the model-organism mechanism plus the human phenotype correlation).



- **Molecular pathways:** chromatin assembly and nucleosome organization; chromatin-templated transcription, replication, repair. No classical signaling cascade (Wnt/MAPK/mTOR) is implicated as primary.
- **Cellular processes:** cell-cycle regulation, DNA-damage repair, transcriptional regulation.

- **Protein dysfunction:** altered histone surface → loss of normal histone–histone/chaperone binding → nucleosome instability (gain-of-function/dominant-negative, not loss of expression).
- **Upstream vs downstream:** the mutation and nucleosome destabilization are **upstream**; transcriptional and cell-cycle/repair defects are **downstream**; the clinical phenotype is the most downstream manifestation.
- **Suggested GO terms:** nucleosome assembly (GO:0006334); chromatin organization (GO:0006325); regulation of transcription by RNA polymerase II (GO:0006357); DNA repair (GO:0006281); cell cycle (GO:0007049); nucleosome (GO:0000786, cellular component).
- **Suggested CL terms:** neuron (CL:0000540); neural progenitor/stem cell (CL:0000047) — affected cell populations during neurodevelopment (inferred).

7. Anatomical Structures Affected

- **Organ/system level:** primarily the **central nervous system** / **brain** (UBERON:0000955), manifesting as intellectual disability and developmental delay. Secondary involvement is variable: **skeletal/growth** (short stature), **craniofacial** (microcephaly, dysmorphism), and occasional structural anomalies.
- **Tissue/cell level:** nervous tissue; neurons and neural progenitors are the inferred principal affected populations, though the molecular lesion is **ubiquitous** (histones are universal), so chromatin dysfunction is cell-non-specific at the molecular level even if the clinical readout is brain-dominant.
- **Subcellular level:** the **nucleus** (GO:0005634) and specifically **chromatin/nucleosome** (GO:0000785/GO:0000786) are the compartments involved.
- **Localization/lateralization:** brain involvement is **bilateral/global**; no lateralized or focal lesion is characteristic.

8. Temporal Development

- **Onset:** congenital/early childhood; developmental delay and microcephaly are apparent in infancy to early childhood.
- **Onset pattern: insidious/chronic** developmental, not acute.
- **Progression: static (non-progressive)** — no neurodegeneration reported, distinguishing it from progressive histonopathies such as H3.3 Bryant–Li–Bhoj syndrome (PMID: 38678163).
- **Course/duration:** lifelong, stable.

- **Critical period:** embryonic/early neurodevelopment is the window of vulnerability (chromatin-dependent developmental programming); no therapeutic window has been defined.

9. Inheritance and Population

- **Inheritance:** autosomal dominant, essentially always **de novo** (sporadic).
- **Epidemiology:** ultra-rare; precise prevalence/incidence are not established. The defining cohort described 29 subjects across six H4 genes, so H4C5-specific counts are small (a handful of published cases).
- **Penetrance:** the neurodevelopmental core appears highly penetrant; non-neurological features show **variable expressivity**.
- **Anticipation / germline mosaicism / founder effects / consanguinity / carrier frequency:** not applicable or not reported — de novo dominant disorders have no carrier frequency and no consanguinity contribution; recurrence risk to siblings is low but not zero owing to possible parental germline mosaicism (general principle; not specifically documented for H4C5).
- **Population demographics:** no ethnic predilection, geographic clustering, or sex bias established; age distribution reflects a pediatric-diagnosed, lifelong condition.

10. Diagnostics

- **Primary diagnostic test: trio whole-exome sequencing (WES) or whole-genome sequencing (WGS)** demonstrating a de novo H4C5 missense variant (PMID: 35202563; PMID: 36987712). Trio testing is essential to establish de novo status and classify pathogenicity (ACMG criterion PS2).
- **Non-diagnostic / exclusionary tests:** single-gene testing is impractical given 14 identical H4 genes; **karyotype, chromosomal microarray, and repeat-expansion testing are non-diagnostic** and serve mainly to exclude mimics (e.g., Angelman syndrome via methylation/UBE3A testing).
- **Biomarkers/imaging/labs:** no specific biomarker, laboratory abnormality, or pathognomonic imaging finding; MRI may show non-specific changes or microcephaly.
- **Emerging:** DNA-methylation **episignature** classification is plausible (proven for several related chromatin disorders — PMID: 38815585; PMID: 42322191) but not yet validated specifically for H4C5.

- **Differential diagnosis: Angelman syndrome** is the key mimic — exome/genome sequencing is recommended for Angelman-like presentations without molecular confirmation (PMID: 36987712); other chromatin-regulator NDDs overlap clinically.

11. Outcome / Prognosis

- **Survival/mortality:** no evidence of reduced life expectancy or disease-specific mortality; it is a non-lethal, non-progressive developmental disorder.
- **Morbidity/function:** principal burden is lifelong intellectual disability and motor impairment with associated disability; variable additional morbidity from seizures, growth, and dysmorphic features.
- **Course:** static; recovery is not expected but neither is deterioration. Supportive therapy can improve functional outcomes.
- **Prognostic factors:** likely related to variant location/severity and degree of intellectual disability; no validated prognostic biomarker. Cancer risk is not elevated to a surveillance-warranting degree (~1% community-wide; PMID: 40241305).

12. Treatment

- **No disease-specific or disease-modifying therapy exists.** Management is **supportive and multidisciplinary**: physical therapy, occupational therapy, speech/language therapy, special education, and developmental support; anti-seizure medication where epilepsy is present; management of growth, feeding, orthopedic, and ophthalmologic issues as they arise.
- **Pharmacogenomics / gene therapy / cell therapy / RNA therapy / targeted therapy / immunotherapy:** none applicable or available.
- **Experimental:** no registered disease-specific clinical trials identified.
- **Suggested NCIT terms:** Supportive Care (NCIT:C19236); Physical Therapy (NCIT:C15917); Occupational Therapy (NCIT:C15214); Speech Therapy (NCIT:C15192); Anticonvulsant Agent (NCIT:C264).

13. Prevention

- **Primary prevention:** none possible for de novo mutations.
- **Secondary prevention:** early developmental intervention to optimize functional outcomes.
- **Genetic counseling:** recurrence risk to future siblings is low (de novo) but not zero due to possible parental germline mosaicism; prenatal or preimplantation testing is feasible for a known familial variant. No population carrier screening is relevant.

- **Cancer surveillance:** not currently recommended (histonopathy cancer incidence ~1%, below the 5% threshold; PMID: 40241305).

14. Other Species / Natural Disease

- **Orthology/conservation:** histone H4 is among the **most evolutionarily conserved proteins** known; H4 genes and the H4K91 interface are conserved from yeast to human, which is why yeast and zebrafish are informative mechanistic models (PMID: 31542297; PMID: 35202563).
- **Natural disease / veterinary relevance:** no naturally occurring H4C5-equivalent disease is documented in companion animals or wildlife (OMIA search not productive for this specific gene). No zoonotic or cross-species transmission is applicable (monogenic disorder).

15. Model Organisms

- **Zebrafish (*Danio rerio*):** embryos expressing H4 disease variants recapitulate abnormal general development, defective head organs, and reduced body-axis length — the principal in vivo disease model demonstrating causality (PMID: 35202563; PMID: 31804630).
- **Yeast / in vitro biochemistry:** nucleosome-reconstitution assays and yeast genetics established that perturbation of H4K91 destabilizes the nucleosome and disrupts transcription, cell cycle, DNA repair, and telomere silencing (PMID: 31542297).
- **Model characteristics:** zebrafish capture gross developmental and craniofacial/CNS phenotypes but do not model higher cognitive deficits; mammalian (mouse) and iPSC/organoid models specific to H4C5 are not yet reported and represent a clear gap.

Mechanistic Model / Interpretation

TEVANED3 is best understood as a **dominant "poison-histone" chromatin disorder**. Unlike loss-of-function NDDs, the pathogenic allele is expressed and its altered protein product is deposited into chromatin, where it corrupts nucleosome architecture from within. Because the mutated residues sit precisely at histone–histone and histone–chaperone interfaces (a statistically notable clustering rather than random distribution), the consequence is **nucleosome destabilization** — directly demonstrated for the recurrent H4K91 interface residue. The downstream ripple effects (transcriptional dysregulation; impaired cell-cycle, DNA-repair, and telomere control) are generic to chromatin instability but become

developmentally catastrophic in the tightly timed, chromatin-dependent programs of neurodevelopment, yielding a penetrant intellectual-disability/developmental-delay core with a variably expressed somatic/craniofacial periphery. The disorder's **static** nature (no neurodegeneration) fits a developmental-programming lesion rather than an ongoing toxic process, and distinguishes it from progressive histonopathies like the H3.3-related Bryant–Li–Bhoj syndrome.

Evidence Base

PMID	Title (abbrev.)	Role in this report
35202563	*Recurrent de novo missense variants across multiple histone H4 genes underlie a neurodevelopmental syndrome*	Defining cohort (n=29); core phenotype; variant clustering at functional interfaces; zebrafish model
36987712	*H4C5 missense variant leads to a neurodevelopmental phenotype overlapping with Angelman syndrome*	Specific H4C5 case (p.Tyr99His, de novo, likely pathogenic); Angelman differential; WES/WGS route
31542297	*Glutarylation of Histone H4 Lysine 91 Regulates Chromatin Dynamics*	Biochemical proof that H4K91 perturbation destabilizes nucleosome and dysregulates transcription/cell cycle/repair
31804630	*A de novo variant in HIST1H4J causes a syndrome analogous to HIST1H4C NDD*	Allelic-series phenotype (growth delay, microcephaly, ID); shared H4 spectrum
40241305	*Deep phenotyping of 192 individuals with germline histonopathies*	Classifies H4 disorders as histonopathies; ~1% cancer incidence → no surveillance
38678163	*Expanded spectrum of Bryant–Li–Bhoj syndrome (H3.3)*	Contrast: H3.3 disorder; supports static vs progressive distinction
38815585; 42322191; 39933316; 42275155	MSL2, SET, KMT2B, DMAP1 NDDs with episignatures	Precedent for episignature diagnostics extendable to H4 disorders
37605493; 34788807	HIST1H1E (Rahman) syndrome	Related linker-histone histonopathy; shared NDD features

Limitations and Knowledge Gaps

1. **Small H4C5-specific sample.** Published H4C5 cases number only a handful; most phenotype data come from the mixed six-gene H4 cohort, so H4C5-specific frequencies, penetrance, and expressivity are imprecise.
2. **No validated episignature** specifically confirmed for H4C5, though strong precedent exists across related chromatin disorders.
3. **No mammalian or iPSC/organoid H4C5 model;** mechanistic inferences for cognition rest on zebrafish, yeast, and biochemistry rather than brain-specific mammalian systems.
4. **Mechanism partly inferred.** Nucleosome destabilization is directly shown for the H4K91 interface; extrapolation to every H4C5 variant (e.g., p.Tyr99His, p.Leu38Pro) is reasonable but not individually demonstrated.
5. **Epidemiology unquantified** (no prevalence/incidence); no formal natural-history or quality-of-life studies.
6. **Ontology gaps:** no dedicated Orphanet/ICD/MeSH codes identified for this serial-numbered subtype in the materials reviewed.

Proposed Follow-up Experiments / Actions

1. **Aggregate an H4C5-specific case series** (GeneMatcher/Matchmaker Exchange) to quantify penetrance, expressivity, and variant-specific phenotype correlations.
2. **Develop and validate a blood DNA-methylation episignature** for H4C5 to reclassify the many VUS in ClinVar.
3. **Functional characterization of individual H4C5 variants** (p.Tyr99His, p.Leu38Pro, p.Arg41Cys) in nucleosome-stability, chromatin-accessibility (ATAC-seq), and transcriptomic assays.
4. **Generate iPSC-derived neurons/organoids and/or a mouse knock-in** to model neurodevelopmental consequences in a mammalian/human context.
5. **Natural-history and quality-of-life study** across histonopathies to define prognosis and care standards.
6. **Confirm/complete cross-ontology mapping** (Orphanet, ICD-11, MeSH) and formalize HPO annotations with frequencies.